

# Notes: Terviö (RES, 2009)

## Superstars and Mediocrities: Market Failure in the Discovery of Talent

### Contents

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Big picture</b>                                                       | <b>2</b>  |
| <b>2</b> | <b>Incremental contribution of the paper</b>                             | <b>2</b>  |
| <b>3</b> | <b>Data and measurement</b>                                              | <b>3</b>  |
| 3.1      | No conventional empirical data . . . . .                                 | 3         |
| 3.2      | Talent, jobs, and revelation . . . . .                                   | 3         |
| 3.3      | Talent distribution and career selection . . . . .                       | 4         |
| 3.4      | Output market and employment . . . . .                                   | 4         |
| <b>4</b> | <b>Models and empirical specifications</b>                               | <b>4</b>  |
| 4.1      | Steady-state workforce composition . . . . .                             | 4         |
| 4.2      | Social planner's threshold . . . . .                                     | 5         |
| 4.3      | Competitive equilibrium wages . . . . .                                  | 5         |
| 4.4      | Unconstrained novice payment . . . . .                                   | 5         |
| 4.5      | Liquidity-constrained equilibrium . . . . .                              | 6         |
| 4.6      | Mediocre types . . . . .                                                 | 6         |
| 4.7      | Long-term wage contracts . . . . .                                       | 6         |
| 4.8      | Risk aversion and quality extensions . . . . .                           | 7         |
| <b>5</b> | <b>Identification and empirical design</b>                               | <b>7</b>  |
| 5.1      | What the model can and cannot identify . . . . .                         | 7         |
| 5.2      | Ideal empirical experiment suggested by the paper . . . . .              | 7         |
| 5.3      | Applications as quasi-identification opportunities . . . . .             | 8         |
| 5.4      | Limitations of the empirical content . . . . .                           | 8         |
| <b>6</b> | <b>Tables and figures</b>                                                | <b>8</b>  |
| 6.1      | Figure 1: Distribution of talent and wages . . . . .                     | 8         |
| 6.2      | Figures 2 and 3: Upside potential and retainment probabilities . . . . . | 9         |
| 6.3      | No empirical tables . . . . .                                            | 10        |
| <b>7</b> | <b>Short summary</b>                                                     | <b>10</b> |
| <b>8</b> | <b>Conclusion</b>                                                        | <b>11</b> |
| 8.1      | Core conclusion . . . . .                                                | 11        |
| 8.2      | Economic intuition . . . . .                                             | 11        |
| 8.3      | Incremental contribution revisited . . . . .                             | 11        |
| 8.4      | Implications for contracts and institutions . . . . .                    | 12        |
| 8.5      | Limitations . . . . .                                                    | 12        |
| 8.6      | Takeaway . . . . .                                                       | 12        |

# 1 Big picture

- **Question.** Why do some occupations generate superstar wages even when there may be many talented people outside the industry? The paper asks whether high talent rents can arise from a market failure in discovering talent, rather than from true scarcity of talent or bad governance.
- **Core idea.** Talent is industry-specific, initially unknown, and revealed only by doing the job. Once revealed, it becomes public information. Firms therefore create information when they try out novices, but the value of good news is later captured by the successful worker as higher wages.
- **Market imperfection.** Workers cannot commit to long-term wage contracts and cannot fully pay upfront for the chance to be tried out. These are ordinary labor-market frictions, but in a talent-discovery market they create underinvestment in experimentation.
- **Main inefficiency.** Firms bid too much for known incumbents and try out too few novices. The industry keeps too many workers whose talent is above the population mean but below the socially efficient rehiring threshold. The paper calls these workers *mediocrities*.
- **Output and wage effects.** Too little experimentation lowers average talent and output. At the same time it raises the price of output and increases wages for known high talents, especially at the top of the distribution.
- **Turnover implication.** Relaxing the worker liquidity or commitment constraint increases the exit rate of young workers, shortens average careers, lowers wages at every known talent level, and lowers top wages the most.
- **Where the problem is severe.** The inefficiency is most severe when talent is hard to predict before actual performance, production costs per job are high, and the right tail of talent is wide. These conditions fit entertainment, top management, entrepreneurship, professional sports, and similar professions.
- **Economic takeaway.** Superstar wages can be a symptom of too little discovery of new talent. The market may pay known stars highly not because talent is fundamentally rare, but because too few people have received real opportunities to reveal their talent.

# 2 Incremental contribution of the paper

- **From efficient superstars to inefficient discovery.** Classic superstar models, especially Rosen-style models, explain high pay as an efficient return to scarce known talent and scale. This paper instead shows that high pay can arise because the market inefficiently discovers too little talent.
- **A public-good view of talent discovery.** Trying out a novice produces information that is valuable to the whole industry. Because successful workers can later sell their revealed talent to all firms, the discovering firm cannot capture the full value of experimentation.
- **A new role for liquidity constraints.** In standard human-capital models, workers cannot always finance training. Here the analogous cost is not training itself but the opportunity cost of giving a scarce job to an untested worker rather than to a known incumbent.

- **A precise definition of mediocrity.** The paper defines mediocre workers as those with talent above the novice prior mean but below the efficient exit threshold. These workers are privately employable but socially should be displaced by novices with more upside potential.
- **Counterintuitive comparative statics.** Policies or institutions that improve novice payment ability or long-term commitment reduce wages and shorten careers. This is very different from standard training models, where better commitment often raises worker-specific investment and lifetime earnings.
- **A link between contract enforceability and inequality.** Enforceable long-term contracts or job-buying by novices would let firms internalize part of the discovery value. This increases experimentation, reduces talent rents, and lowers wage dispersion.
- **Separation from the classic superstar effect.** The paper later lets talent determine quality rather than quantity. The mediocrity effect remains even when Rosen’s superstar mechanism is present, so the two mechanisms are conceptually distinct but can reinforce each other.
- **Applied implications.** The paper maps the theory to Hollywood’s studio system, record deals, professional team sports, and high-level managerial labor markets. These examples illustrate how contract length and enforceability can affect discovery, turnover, and superstar rents.

### 3 Data and measurement

#### 3.1 No conventional empirical data

- This is a theory paper. It does not build an archival data set or estimate regressions.
- The paper’s “measurement” problem is conceptual: how should the model measure talent, experimentation, career length, output, wages, and the cost of trying out new workers?
- The paper uses figures to illustrate the model’s equilibrium and comparative statics rather than tables of empirical estimates.

**Interpretation:** The paper should be read as a mechanism paper. Its goal is to show that ordinary labor-market imperfections can produce inefficient talent discovery and superstar rents.

#### 3.2 Talent, jobs, and revelation

- Each firm combines one worker with other inputs at cost  $c > 0$  per period.
- Output equals the worker’s talent, denoted  $\theta$ .
- There is an unlimited supply of potential workers with outside wage  $w_0 > 0$ .
- A novice’s talent is unknown to everyone, including the novice.
- Talent is industry-specific and becomes public after one period of work.
- After talent is revealed, a retained worker can work in the industry for  $T$  more periods.
- Workers and firms are risk neutral in the baseline model, and there is no discounting.

**Interpretation:** The key assumption is public learning. A firm that discovers a star cannot keep that information private. Other firms can bid for the worker once talent is revealed.

### 3.3 Talent distribution and career selection

- Talent is drawn from a continuous distribution  $F$  with positive support  $[\theta_{min}, \theta_{max}]$ .
- Let  $\bar{\theta}$  denote the population mean talent of novices.
- In any steady state, careers are summarized by a rehiring threshold  $\psi$ .
- Workers with revealed talent below  $\psi$  exit the industry after one period.
- Workers with revealed talent above  $\psi$  remain for  $T$  more periods.

**Interpretation:** The exit threshold is the central object of the paper. A higher threshold means more experimentation with novices, higher average talent, more turnover, and fewer mediocre incumbents.

### 3.4 Output market and employment

- The industry faces a downward-sloping demand curve  $p^d(q)$ .
- The mass of firms and jobs is a continuous variable  $I$ , equal to industry employment.
- The product price adjusts so that novice-hiring firms break even in competitive equilibrium.
- Long-term wage contracts are not enforceable in the baseline because workers cannot commit to reject higher future offers.

**Interpretation:** The competitive product market is important. When the workforce contains too many mediocre incumbents, average talent is too low and output is costly. The output price rises to make novice hiring break even.

## 4 Models and empirical specifications

### 4.1 Steady-state workforce composition

Let  $i(\psi)$  be the fraction of novices in the workforce when the exit threshold is  $\psi$ . Flow balance implies

$$i(\psi) = \frac{1}{1 + T(1 - F(\psi))}. \quad (1)$$

The average talent in the workforce is

$$A(\psi) = i(\psi)\bar{\theta} + (1 - i(\psi))E[\theta \mid \theta > \psi]. \quad (2)$$

Substituting the expression for  $i(\psi)$  gives

$$A(\psi) = \frac{1}{1 + T(1 - F(\psi))}\bar{\theta} + \frac{T(1 - F(\psi))}{1 + T(1 - F(\psi))}E[\theta \mid \theta > \psi]. \quad (3)$$

**Interpretation:** A higher threshold improves the quality of retained veterans but also increases the share of novice workers. The socially efficient threshold balances current output losses against the option value of discovering high talent.

## 4.2 Social planner's threshold

The planner chooses the threshold to maximize average talent. The first-order condition can be written as

$$\psi - \bar{\theta} = T(1 - F(\psi)) (E[\theta \mid \theta > \psi] - \psi). \quad (4)$$

Let  $A^*$  denote the solution. The paper shows that the maximized average talent equals the optimal exit threshold:

$$A^* = A(A^*). \quad (5)$$

- The left-hand side is the current output cost of using a novice instead of the marginal retained veteran.
- The right-hand side is the expected future gain from discovering a worker above the threshold.
- The optimal threshold is unique and lies above the population mean  $\bar{\theta}$ .

**Interpretation:** Efficient experimentation is deliberately harsh. Some workers who are better than the average novice should still be replaced because a new novice has upside potential.

## 4.3 Competitive equilibrium wages

Given an equilibrium threshold  $\psi$ , the zero-profit condition for firms implies

$$P(\psi) = \frac{w_0 + c}{\psi}, \quad (6)$$

where  $P(\psi)$  is the product price. The wage of a worker with revealed talent  $\theta$  is

$$w(\theta \mid \psi) = (w_0 + c) \left( \frac{\theta}{\psi} - 1 \right) + w_0. \quad (7)$$

- The threshold worker earns exactly the outside wage  $w_0$ .
- Higher talent workers earn rents because their output exceeds the threshold type's output.
- A lower threshold raises the wage slope and increases top talent rents.

**Interpretation:** The wage function is a simple Ricardian rent schedule. When the threshold is inefficiently low, every retained worker is compared to a weaker marginal worker, and output price is higher. Both channels raise talent rents.

## 4.4 Unconstrained novice payment

If novices can fully pay for the chance to reveal their talent, the competitive equilibrium is efficient. The required payment is

$$b^* = P^*(A^* - \bar{\theta}) = (w_0 + c) \left( 1 - \frac{\bar{\theta}}{A^*} \right). \quad (8)$$

- $b^*$  is the efficient job price.
- It compensates the firm for replacing a known threshold worker with a lower-expected-output novice.
- The payment can be large when production costs  $w_0 + c$  are high or when novice upside potential is high.

**Interpretation:** The financing need is not just the worker’s own wage. If the job requires expensive complementary inputs, the novice would need to finance a fraction of the full production cost.

#### 4.5 Liquidity-constrained equilibrium

Suppose novices can pay at most  $b < b^*$ . Then the novice wage cannot fall enough to make firms internalize the option value of experimentation. The equilibrium threshold becomes

$$\psi(b) = \begin{cases} \frac{(w_0 + c)\bar{\theta}}{w_0 + c - b}, & b < b^*, \\ A^*, & b \geq b^*. \end{cases} \quad (9)$$

- If  $b = 0$ , the threshold is the population mean  $\bar{\theta}$ .
- As  $b$  rises, the threshold rises toward the efficient threshold  $A^*$ .
- Low  $b$  implies too many retained veterans and too few novices.

**Interpretation:** Liquidity constraints make the market too lenient toward incumbents. The industry retains workers who are privately profitable at the distorted price but socially inefficient to keep.

#### 4.6 Mediocre types

The paper defines mediocre types as

$$\theta \in (\bar{\theta}, A^*). \quad (10)$$

- These workers are better than the average novice.
- They are nevertheless below the efficient rehiring threshold.
- They should be displaced by novices in the efficient allocation.
- Under liquidity constraints, many of them remain in the industry.

**Interpretation:** “Mediocre” does not mean low ability in an absolute sense. It means socially inefficiently retained relative to the option value of experimenting with new talent.

#### 4.7 Long-term wage contracts

The paper then allows workers to commit to an initial long-term wage contract for  $1 + S$  periods. This gives firms a share of the rents from talents they discover.

- Longer commitment raises the rehiring threshold.
- Wages of free agents fall at all talent levels.
- Top wages fall the most.
- Turnover increases and average careers shorten.
- Full efficiency requires commitment for the full career length.

**Interpretation:** Contract commitment is another way for workers to pay for discovery. By letting firms capture some future rents, it induces more experimentation with novices.

## 4.8 Risk aversion and quality extensions

- With risk-averse individuals, even financially unconstrained novices do not want to pay the risk-neutral value of the talent-discovery option.
- Greater risk aversion lowers the equilibrium threshold and raises wages for all talent levels.
- When talent is modeled as quality rather than quantity, the same core mechanism survives.
- In the quality version, the constrained equilibrium also distorts output per worker upward: each type produces more output than in the efficient allocation.

**Interpretation:** The baseline liquidity constraint is not the only way to generate the inefficiency. Incomplete insurance against one's own unknown talent can also make novices unwilling to finance efficient experimentation.

## 5 Identification and empirical design

### 5.1 What the model can and cannot identify

- The paper is not an empirical identification paper. It does not estimate a causal treatment effect in data.
- It identifies a theoretical mechanism: public on-the-job talent revelation plus limited worker commitment or liquidity leads to inefficiently low experimentation.
- The model generates comparative statics that could be tested in settings with exogenous changes in contract enforceability or worker ability to finance entry.

**Interpretation:** The strongest claim is conceptual: superstar rents can come from inefficient discovery. Empirical validation would require a natural experiment that changes the friction behind the inefficiency.

### 5.2 Ideal empirical experiment suggested by the paper

The paper emphasizes that a useful natural experiment would change either:

- workers' ability to pay for jobs or accept low initial wages;
- enforceability of long-term wage contracts;
- rules governing non-compete clauses, reserve clauses, loan-outs, or similar contractual restrictions;
- institutional mechanisms that let firms capture returns to discovering talent.

The predicted effects of relaxing the friction are:

- more novices tried out;
- higher turnover among young workers;
- shorter average careers;
- lower wages for incumbents of every known talent level;
- a larger decline in the highest wages;

- higher average talent and output.

**Interpretation:** The empirical signature is unusual: an efficiency-enhancing change can lower superstar wages and increase early-career exit. That makes the paper’s predictions distinguishable from many standard training or human-capital models.

### 5.3 Applications as quasi-identification opportunities

- **Hollywood studio system.** Long-term contracts in old Hollywood allowed studios to capture rents from actors whose talent they discovered. The breakdown of the studio system should have increased veteran pay, increased dispersion, and reduced experimentation.
- **Record deals.** Multi-album contracts let record companies capture part of the payoff from discovering musicians. Weakening these contracts should reduce new-artist experimentation and raise rents for proven artists.
- **Professional team sports.** Restrictions on player mobility can partly function as mechanisms that let teams capture returns from player development. Changes in contract duration or renegotiation rights can therefore alter discovery incentives.
- **Executive and professional markets.** The model is relevant when performance is publicly observable, pre-job prediction is poor, and real opportunities are scarce or costly.

**Interpretation:** The model points to institutional variation rather than cross-sectional wage data alone. Wage patterns are not enough because high wages may reflect either scarcity of true talent or scarcity of revealed talent.

### 5.4 Limitations of the empirical content

- The paper abstracts from many real labor-market features, including search costs, asymmetric information, effort problems, career concerns, and firm-specific training.
- The one-shot learning process is a simplifying assumption; talent may often be revealed gradually.
- The model does not explicitly model rent dissipation through queues, unpaid effort, low-probability entry, or willingness to bear risk.
- It is difficult to infer the welfare loss from observed wage data without observing the counterfactual pool of unrevealed talent.

**Interpretation:** The model is deliberately stark. Its value is to isolate a force that can be hidden in ordinary wage data: too many incumbents and too little experimentation with novice talent.

## 6 Tables and figures

### 6.1 Figure 1: Distribution of talent and wages

Read:

- The top panel plots talent in the workforce by quantile under three regimes: unconstrained, fully constrained, and an intermediate case.



- The black line is the unconstrained case. It starts at the efficient threshold  $A^*$ , meaning mediocre types below  $A^*$  have been displaced by novices.
- The dark grey line is the fully constrained case, with a lower threshold near the population mean  $\bar{\theta}$ .
- The bottom panel plots the wage schedule. Lower thresholds create steeper wage schedules and much higher wages for high-talent workers.
- The figure uses the example  $c = 100$ ,  $w_0 = 10$ ,  $T = 15$ , and a uniform  $[0, 1]$  talent distribution.

**Economic message:** The same friction creates two outcomes at once: too many mediocre incumbents and too much rent for known stars. Reducing the friction raises the quality threshold and lowers wage dispersion.

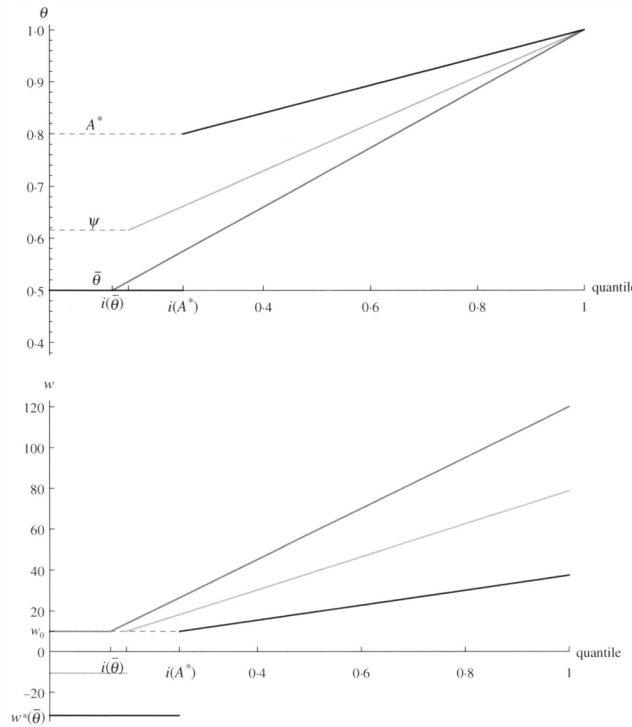


FIGURE 1  
Distribution of talent (top panel) and wages (bottom panel) in the workforce. Black lines refer to the unconstrained case ( $b \geq b^*$ ), dark grey to the fully constrained case ( $b = 0$ ), and light grey to a middle case ( $b = b^*/2$ ). The graphs were drawn assuming  $c = 100$ ,  $w_0 = 10$ ,  $T = 15$ , and talent drawn from the uniform  $[0, 1]$  distribution. The efficient threshold is then  $A^* = \frac{\sqrt{1+T}}{1+\sqrt{1+T}} = 0.8$ .

Figure 1: Distribution of talent and wages in the workforce.

## 6.2 Figures 2 and 3: Upside potential and retainment probabilities

Read:

- Figure 2 plots the novice productivity shortfall relative to the efficient threshold type under Weibull talent distributions.
- The shortfall is larger when talent is more right-skewed and when the potential veteran career length  $T$  is larger.

- A large shortfall means novices would need to finance a large fraction of production costs to induce efficient experimentation.
- Figure 3 plots the efficient proportion of novices retained under the same distributional assumptions.
- More right-skewed talent distributions imply that only a small share of novices should be retained efficiently.
- The dashed line in Figure 3 shows the constrained case  $b = 0$ , where retention does not adjust enough to the upside potential of experimentation.

**Economic message:** The model is most powerful in superstar-like environments: talent has a long right tail, only a few novices should be retained, and the efficient job price can be a large share of total production costs. Small entry fees or small wage discounts are not enough to fix the problem.

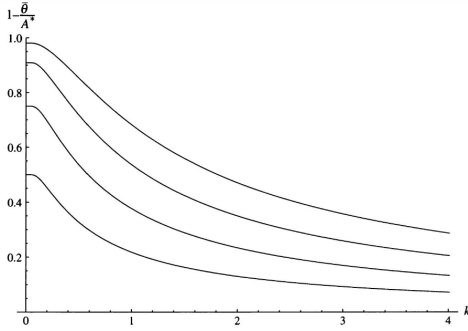


FIGURE 2  
Shortfall of novice productivity relative to efficient threshold type, when  $\theta \sim \text{Weibull}(k, \lambda)$  and  $T = 1, 3, 10, 50$  (from bottom to top), any  $\lambda > 0$

Figure 2: Shortfall of novice productivity relative to the efficient threshold.

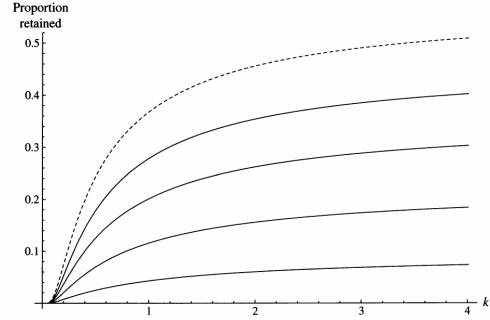


FIGURE 3  
Solid lines show the efficient proportion of novices retained, when  $\theta \sim \text{Weibull}(k, \lambda)$  and  $T = 1, 3, 10, 50$  (from top to bottom), any  $\lambda > 0$ . Dashed line gives the proportion retained in the constrained case ( $b = 0$ ) for any  $T$

Figure 3: Efficient and constrained retainment probabilities.

### 6.3 No empirical tables

- The paper does not contain empirical regression tables.
- The main quantitative objects are theoretical equations, propositions, and the three figures reproduced above.
- The figures summarize the core comparative statics: constrained markets retain too many incumbents, raise top wages, and make experimentation most valuable when talent has high upside potential.

**Economic message:** The absence of tables reflects the paper's role as a theory contribution. The empirical value lies in the sharp predictions it gives for future natural experiments.

## 7 Short summary

- The paper studies labor markets where talent is industry-specific, initially unknown, and publicly revealed through actual job performance.

- Firms underinvest in trying out novices because successful discoveries become public and the worker captures future rents.
- If novices could pay the efficient job price, the competitive equilibrium would be efficient.
- If novices cannot pay enough or cannot commit to long-term wages, the industry retains too many mediocre incumbents.
- Mediocre incumbents are workers whose talent is above the novice mean but below the efficient rehiring threshold.
- The constrained equilibrium has lower average talent, lower output, higher output price, higher wage dispersion, and higher wages for known stars.
- Relaxing liquidity or commitment constraints raises turnover, shortens careers, and lowers wages, especially for the highest talents.
- Risk aversion creates a similar inefficiency because novices are unwilling to pay the risk-neutral value of a risky talent-discovery option.
- The results are robust when talent is modeled as quality rather than quantity.
- The model applies most naturally to entertainment, entrepreneurship, top management, sports, and other markets where real opportunities to reveal talent are scarce and costly.

## 8 Conclusion

### 8.1 Core conclusion

Terviö shows that superstar wages can arise from a market failure in the discovery of talent. When talent is revealed only through actual performance and then becomes public information, firms do not have enough incentive to experiment with untried workers. If workers cannot buy entry opportunities or commit to low long-term wages, the industry underinvests in discovery.

### 8.2 Economic intuition

The central trade-off is between current output and information. A known mediocre incumbent may have higher expected output today than a novice. But the novice has upside potential: if the novice turns out to be a star, the industry gains a high-talent worker for many future periods. The social planner values this option. A competitive firm does not fully value it because the worker can later sell the revealed talent to other employers.

### 8.3 Incremental contribution revisited

The paper changes the interpretation of superstar wages. High rents to known stars need not imply that talent itself is scarce. They can imply that opportunities to reveal talent are scarce. The paper therefore complements and challenges classic superstar economics. It accepts that high-talent workers can generate large rents, but asks why so few high talents are discovered in the first place.

## 8.4 Implications for contracts and institutions

The model gives an efficiency rationale for institutions that look restrictive at first glance, such as long-term contracts, reserve clauses, or development systems. These institutions can let firms capture some returns from discovering talent and thereby encourage more experimentation. At the same time, such institutions reduce incumbent rents and wage inequality, which creates a conflict between efficiency and the interests of successful incumbents.

## 8.5 Limitations

The model is intentionally simple. It abstracts from search frictions, heterogeneous firms, effort, private information, on-the-job training, and gradual learning in the main analysis. It also does not quantify the welfare loss in a real market. Empirical measurement would require a natural experiment that changes contract enforceability or novice financing ability while leaving talent demand otherwise comparable.

## 8.6 Takeaway

The takeaway is that the labor market may be bad at discovering talent precisely in the occupations where talent appears most publicly visible. Public performance makes successful talent easy to bid away, which weakens firms' incentives to create discovery opportunities. The result can be a market with too many mediocrities, too few new chances, and very high pay for the limited pool of proven stars.